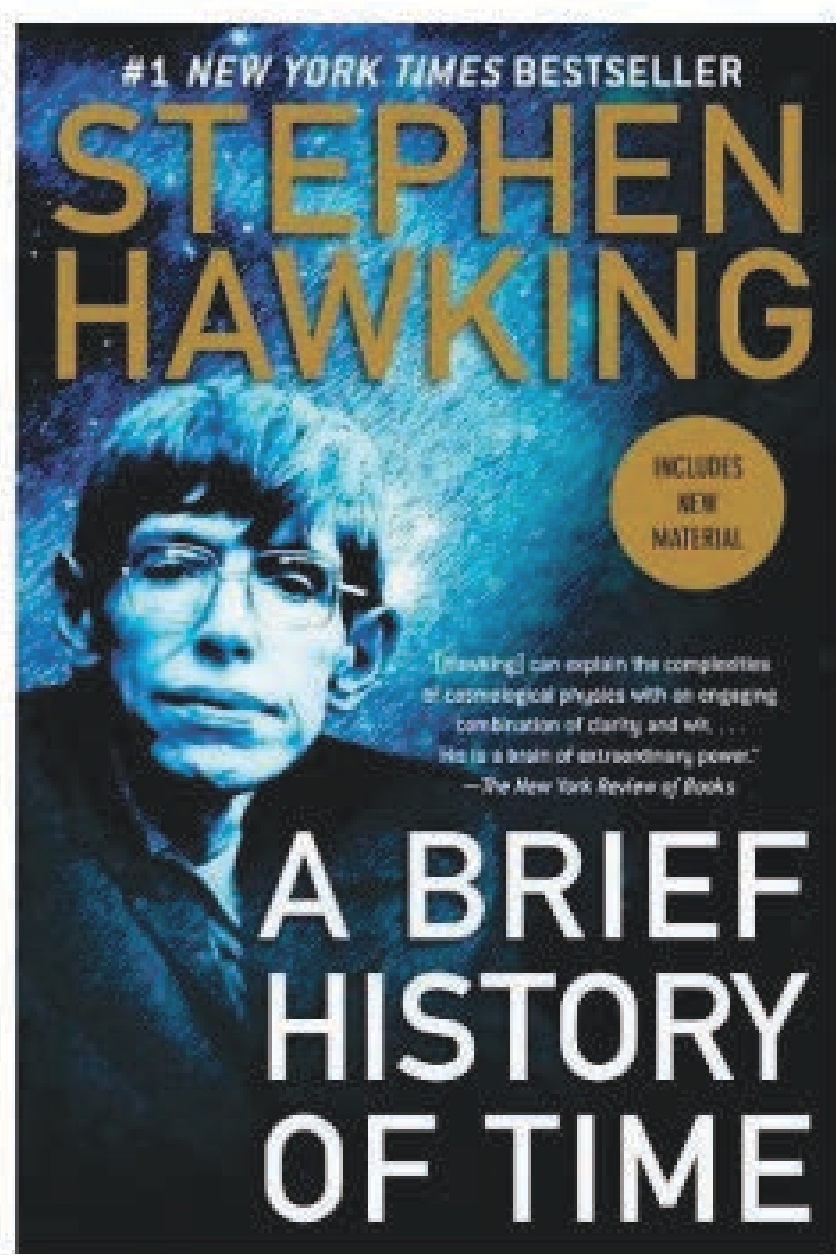




A BRIEF HISTORY OF TIME

STEPHEN HAWKING

The Big “So What”



Mankind has always sought to understand our world. We want the answers to questions like: When/how was universe created? Where did we come from? Where are we going? Using simple layman terms, this book explains abstract concepts about cosmology—such as space, time, black holes, quantum theory and the expanding universe—to help us understand our universe in scientific terms.

Introduction



To have a meaningful discussion of the universe, we must first understand what a scientific theory is.

- A good theory must (i) accurately describe events (with rare exceptions) based on a model, and (ii) accurately predict future outcomes or observations.
- However, all theories are provisional because something that has been true so far can still be contradicted in the future. All it takes is 1 false event to disprove a theory. We can have more confidence in a theory if it makes accurate predictions consistently, but we can never prove a theory definitively.



While the universe may seem vast and unknowable, it actually follows certain laws that can be observed, understood and predicted. Over the centuries, mankind has vastly updated our knowledge by making observations, forming theories, then constantly testing, discarding or refining them based on how well our observations matched our predictions.

KEY QUOTES

“A theory is just a model of the universe, or a restricted part of it, and a set of rules that relate quantities in the model to observations that we make.”

“Any physical theory is always provisional, in the sense that it is only a hypothesis: you can never prove it.”

- For instance, Aristotle first theorized in 340 B.C. that Earth was round (not flat) and that the sun, moon, stars and planets moved around Earth in circular orbits. Many different theories were debated thereafter. In the 1600s, our predictions finally aligned with our observations, when it was theorized that the moon, stars, and planets (including Earth) orbited around the sun in ellipses (elongated circles). Then in 1687, Newton's theory on gravity explained *why* this happened—the celestial bodies were being attracted to one another by gravitational forces.
- Today, we still have an incomplete understanding of our universe. We can explain and predict parts of it through separate theories, but we lack a unified theory that fully explains the whole universe. This is the ultimate goal of science and our quest for knowledge.

KEY QUOTES

"Today we still yearn to know why we are here and where we came from...And our goal is nothing less than a complete description of the universe we live in."

What We Know about our Universe



Everything is relative. Nothing is absolute.

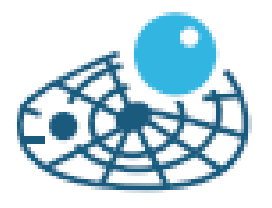


Everything is in motion. In the past, people assumed that an object's natural state was at rest, i.e. it'd stay totally immobile if there was no force acting on it. Sir Isaac Newton disproved this with his Laws of Gravity. The laws state that everything is *already* in motion and objects are constantly moving in relation to one another. A stationary chair may *seem* to be at rest, but it's in motion since it's sitting on a moving Earth. Specifically:

- The first law says that an object will keep moving in a straight line provided it is not acted on by another force, e.g. balls will roll down a slope in a straight line if gravity is the only force.
- The second law says that an object will accelerate at a rate proportional to the force acting on it, e.g. a car with a strong engine will accelerate faster than one with a

weak engine. And, the greater the body's mass, the less its motion is affected by the force, i.e. a heavier car takes longer to accelerate.

- Newton's third law says that all bodies attract other bodies with a force proportional to the mass of each object.



Space and time are not absolute; they are relative to the observer.

- Space is relative, i.e. there's no absolute standard of space. Since everything is constantly moving relative to everything else, it's impossible to determine if 2 events occurring at different times actually happened in the same space.
- Time is also relative, i.e. it moves at different speeds for different observers.
 - (i) Ole Rømer discovered that light moves at a uniform speed (186,000 miles/second) regardless of the observer's position or speed of travel. But, Newton's law says that nothing is fully at rest, so light must travel at a constant speed relative to *something*. Speed is calculated as the distance travelled divided by time ($S = D/T$). If you drive at 100km/hr, you'll be 100km closer to your destination after 1 hour. But if you travel at the speed of light which is *always* the same, S is constant as D changes, which means T must also be changing.
 - (ii) Albert Einstein and Henri Poincaré separately and independently concluded that time slows down as you approach the speed of light. Einstein's equation $E=mc^2$ (where E = energy, m = mass and c = speed of light) says that an object's mass increases as its energy and speed increase. Objects can approach the speed of light but never actually reach it, and nothing can travel faster than the speed of light.



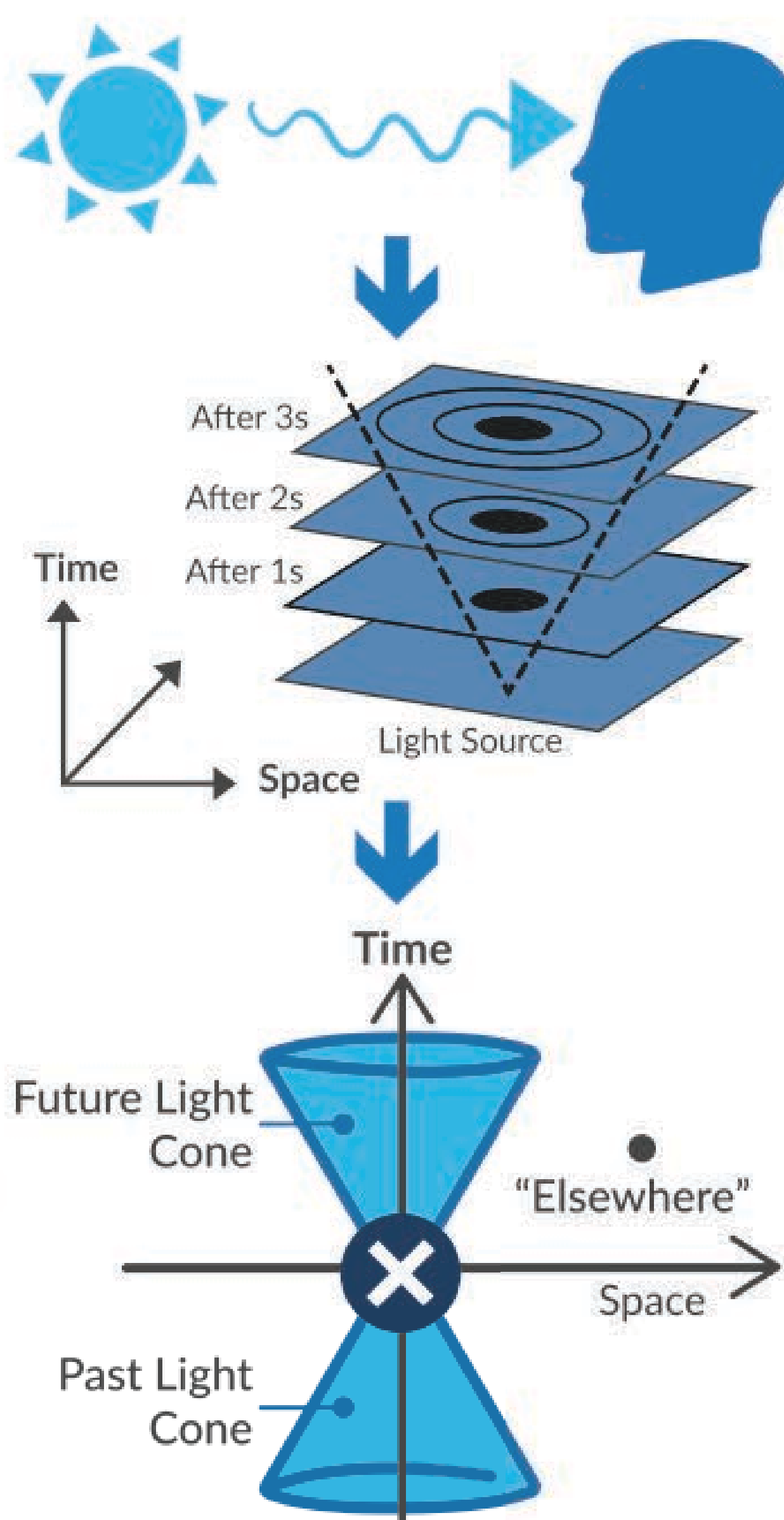
As we've seen, space and time can't be treated separately. They are interwoven into the fabric of space-time.

KEY QUOTES

"The theory of relativity... [forces] us to change fundamentally our ideas of space and time. We must accept that time is not completely separate from and independent of space, but is combined with it to form an object called space-time."

- All events must be mapped in space-time relative to 4 coordinates: (a) a point in time and (b) a point in space, relative to (c) the observer's position and (d) the observer's motion.

- Light moves in waves and expands in a widening curve as it travels through space-time, just like how a rock creates ripples in a pond. You can visualize the effect as a light-cone. The audience in the future light cone can see Event X. Event X in turn can receive signals from those in the past light cone, and possibly be influenced by them. Events that lie outside the future/past light cones are said to lie in the event's "elsewhere"—they neither affect nor are affected by what happens at X.



- Let's imagine that Event X is the sun burning out. Since the sun is so far away, it takes time for the light-waves to reach us. By the time we see the sun burning out, the event is long past. So when you look at the starry sky, you're seeing how the universe looked like in the past, i.e. you're in the future light cone of the light from those stars.



In 1916, Einstein's general theory of relativity added gravity to the picture. Einstein theorized that space-time isn't flat and doesn't travel in a straight line.

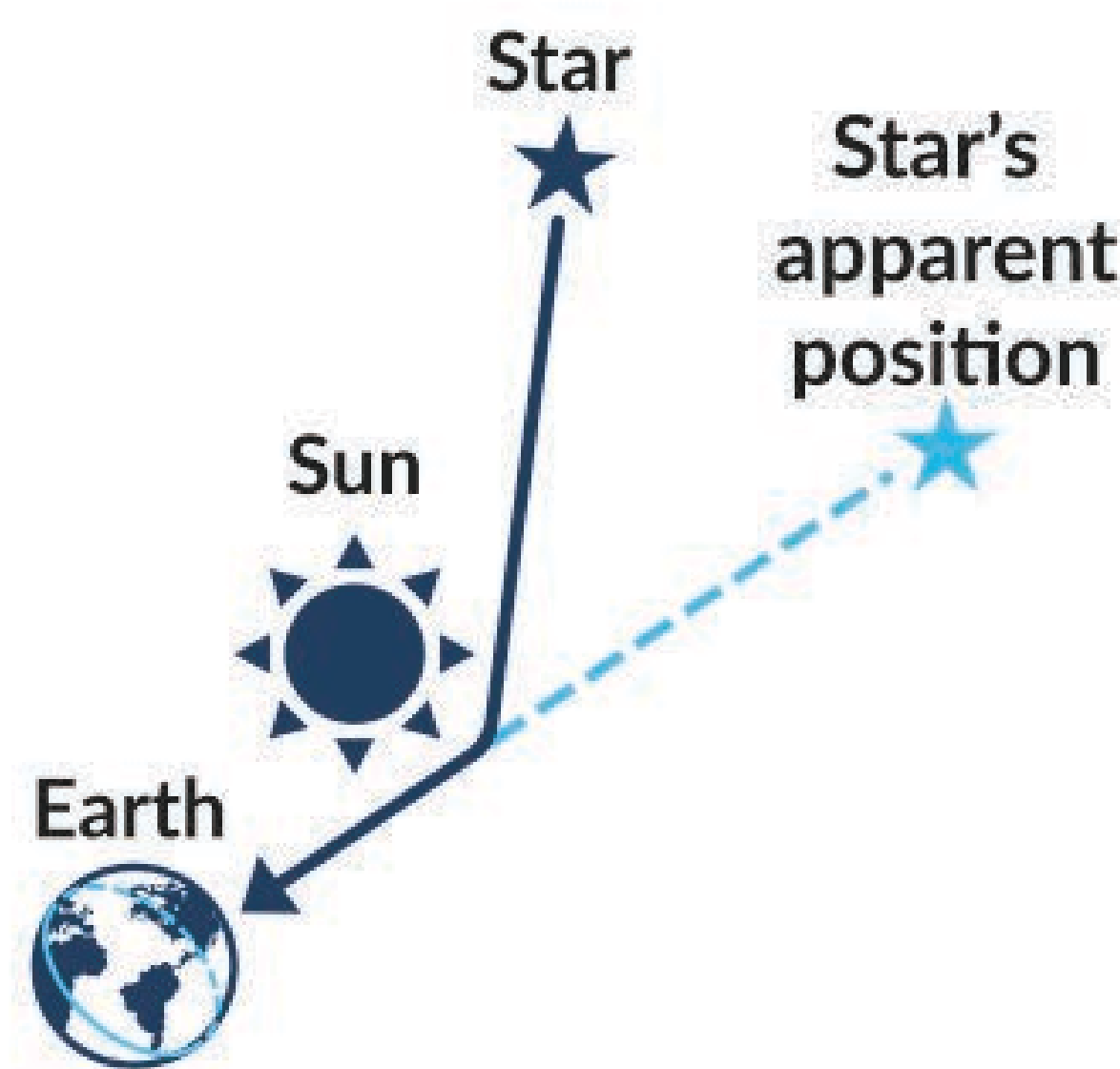
- Imagine putting a heavy object on a stretched-out piece of fabric—the weight of the object will cause the fabric to sink. Likewise, gravity warps the curvature of space-time so it travels with a slight curve instead of a straight line. Due to the space-time curvature, distant stars actually appear to be in a different position than their actual location.

KEY QUOTES

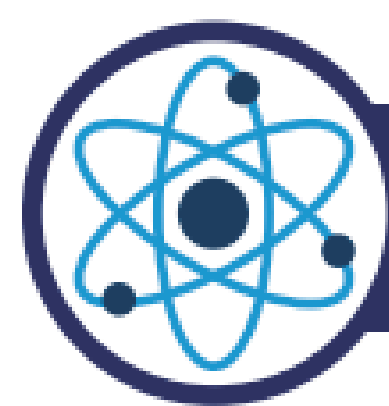
"Space and time not only affect but also are affected by everything that happens in the universe."

"When we look at the universe, we are seeing it as it was in the past."

"In the theory of relativity there is no unique absolute time, but instead each individual has his own personal measure of time that depends on where he is and how he is moving."



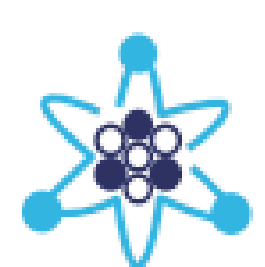
- You'll experience time differently depending on where you are and how you're moving. If you're travelling in space at close to the speed of light with a low gravitational pull, 50 years later you'll look much younger than your twin who stayed on Earth.
- Einstein's general theory of relativity explains most things in the universe. Yet, it is incomplete because it can't explain how the universe was started. This is where quantum theory comes in.



The Uncertainty Principle and Quantum Mechanics



For a long time, scientists believed in determinism, i.e. we'd be able to predict everything so long as we had perfect and complete info. However, this was quashed in 1926 when scientist Heisenberg presented the uncertainty principle. To measure the position and velocity of a particle, you must shine a high-frequency light on it. This would disturb the particle and disrupt its velocity in unpredictable ways. So, it's impossible to accurately observe and gauge a particle's position and speed. If we can't precisely measure the state of the universe at any point in time, then we can't accurately predict the future.



This laid the foundation for the new field of quantum mechanics:

- Quantum mechanics recognizes randomness and adopts a probabilistic approach. Instead of trying to predict exactly how particles will move to get 1 definitive result, it predicts the probabilities of various possible outcomes.

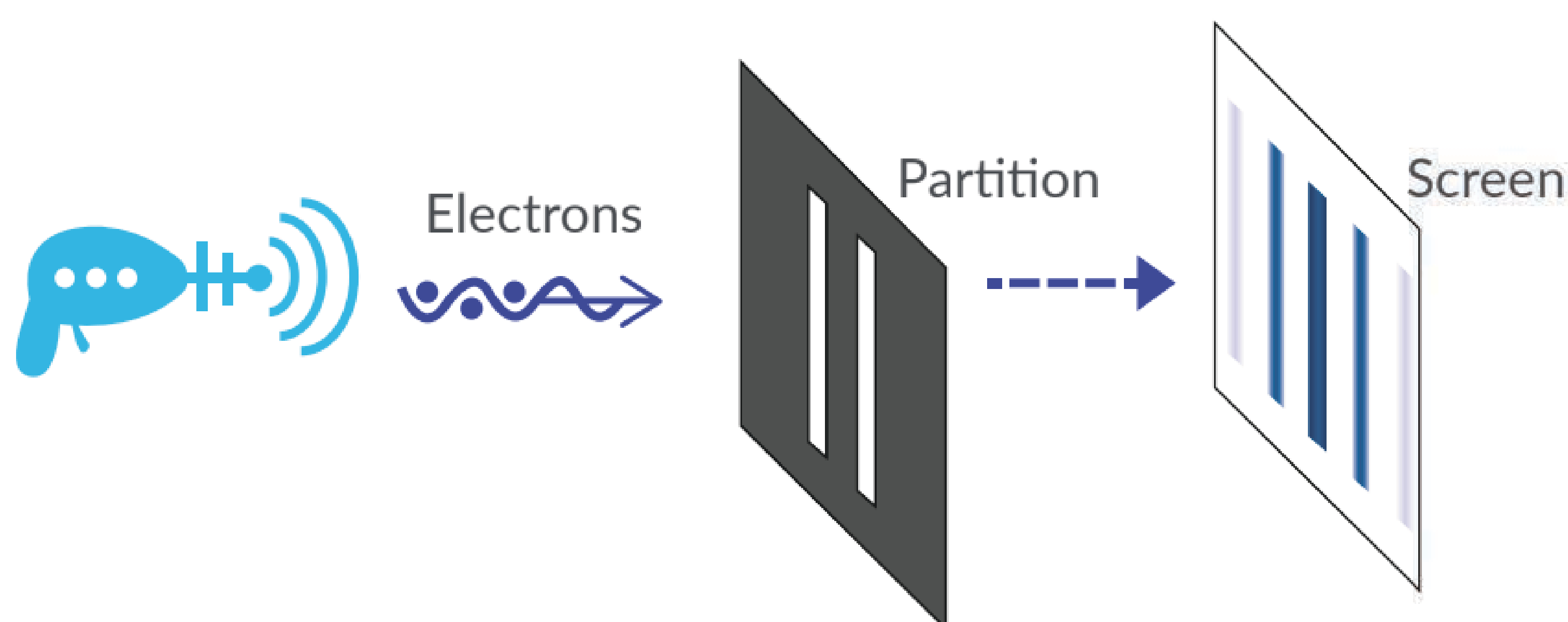
KEY QUOTES

"We have seen...how, in less than half a century, man's view of the universe, formed over millennia, has been transformed."

"One certainly cannot predict future events exactly if one cannot even measure the present state of the universe precisely!"

"One could predict the approximate number of times that the result would be A or B, but one could not predict the specific result of an individual measurement."

- It also looks at the physical world in terms of both particles (e.g. atoms) and waves (e.g. light or radio waves). This is significant because there may be interference when 2 sets of waves or particles coincide—they may (i) reinforce each other to form stronger waves or (ii) cancel each other out.
- The two-slit experiment shows that light/matter can display the characteristics of both waves and particles. When you shine a light-source through a partition with 2 parallel slits, you'll see bright-and-dark bands on the screen behind—this shows the wave characteristics of light. Even if the electrons are sent one at a time, you'll still get the fringes instead of a uniform distribution, i.e. each electron is passing through both slits concurrently.



The Building Blocks of Nature



To understand how the universe works, we must understand its building blocks. Quarks are the **smallest known bits in the universe that are believed to make up everything.**

- Atoms are the basic building blocks of matter, and can be combined to create molecules. By the early 20th century, we know that each atom is made of negatively-charged electrons orbiting around a positively-charged nucleus (made of positive protons and neutral neutrons).
- In the 1960s, scientists discovered that protons and neutrons are made up of even smaller particles—quarks.
 - (i) A quark's "*spin*" describes how it looks from different angles (e.g. a particle that looks the same from any direction has a spin of 0).

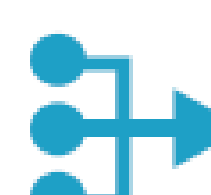
KEY QUOTES

"We do have some theoretical reasons for believing that we have, or are very near to, a knowledge of the ultimate building blocks of nature."

- (ii) The *exclusion principle* says that 2 similar particles can't have the exact same position and velocity.
- (iii) Every particle has an *antiparticle* that can annihilate each other (e.g. electron vs positron). The existence of observable matter means that the universe has more quarks (and the protons/neutrons they form) than antiquarks, antiprotons, and antineutrons.

 Matter particles interact via force-carrying particles.

- A matter particle (e.g. an electron) can emit a force-carrying particle. This creates a recoil that changes the velocity of the matter particle. The force-carrying particle then collides with another matter particle to change the latter's velocity. With repeated collisions, they can create strong forces.
- There are 4 types of force-carrying particles, each with different force-levels and nature of interactions: (i) *gravitons* carry gravitational forces, (ii) *photons* produce electromagnetic forces, (iii) *bosons* create weak nuclear forces and (iv) *gluons* bind quarks to protons and neutrons to create the nucleus in atoms.

 The grand unified theory (GUT) says that at extremely high energies, nuclear forces can be weakened to unbind gluons and quarks so they act like free agents. At such energy levels, it's theoretically possible to merge electromagnetic, weak, and strong forces into a single force. However, there's currently no conceivable way to generate that level of energy. The GUT also predicts that it's possible for protons to randomly decay into particles like antielectrons, though this phenomenon has yet to be observed.



The Big Bang and the Expanding Universe

 For the bulk of human history, people assumed that the universe either (i) had always existed in an infinite, unchanging state, or (ii) was created at a finite point in time (by God or a Creator in line with religious and theological beliefs). However, we later discovered that the universe is actually dynamic and expanding.

KEY QUOTES

- Light is made up of the spectrum of component colors found in a rainbow. The Doppler Effect says that the frequency of sound/light/other waves changes as the source moves toward or away from the observer. The wavelength of the waves you receive is longer if the source is moving *away* from you (called “red-shift” since the longest wavelengths are at the red end of the spectrum). The wavelength is shorter if the source is moving *toward* you (called “blue-shift”). That’s why the engine sounds of a car approaching you seems to have a higher pitch compared to one that’s driving away from you.
- By studying wavelengths of visible light, astronomers found that most galaxies are red-shifting or moving away from us. The universe is expanding at a rate fast enough to overcome the forces of gravity (which would otherwise have caused it to contract). This is similar to how a rocket must travel fast enough to escape Earth’s gravitational pull to enter orbit.

✦ If the universe has been expanding for eons, then there must have been a time when it was infinitesimally small. That’s the Big Bang.

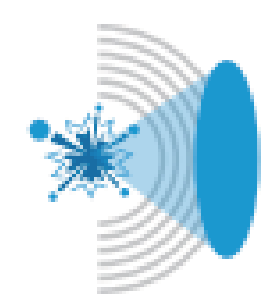
- For the bulk of the 20th century, the “Hot Big Bang” model was used to explain how our universe came to be. According to this model, the universe was incredibly hot right after the Big Bang explosion. Particles moved rapidly to form protons, neutrons, atoms, and molecules. Within seconds, the universe had expanded, started to cool, and particles slowed down to form new elements through a series of nuclear fusions. Within a few hours, most of the hydrogen and helium in our universe today were formed in huge clouds. Over millions of years—as the universe kept expanding and cooling—these became the building blocks of new atoms and molecules that eventually made up all other matter (including Earth and humans).
- This model explains much of our observable universe, but not all of it. One of the unanswered questions was: why/how did the universe expand so uniformly across space-time on a large scale? It’s tempting to use theology to explain away such unknowns—maybe God simply

KEY QUOTES

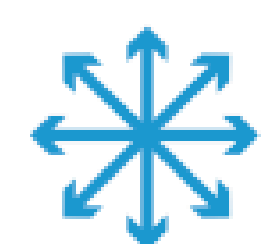
“The old idea of an essentially unchanging universe that could have existed, and could continue to exist, forever was replaced by the notion of a dynamic, expanding universe that seemed to have begun a finite time ago, and that might end at a finite time in the future.”

“So long as the universe had a beginning, we could suppose it had a creator. But if the universe is really completely self-contained, having no boundary or edge, it would have neither beginning nor end...What place, then, for a creator?”

decided on those conditions for His divine purposes. Yet, scientists continue to search for concrete laws to explain these observations.



Guth's inflationary theory helps to explain this strange uniformity. This theory says that after the Big Bang, the universe ballooned to a million million million million times its radius in <1 second. This rate of expansion was much faster than the Hot Big Bang model, and would've caused the universe to supercool so fast that forces would stay unified instead of decoupling. The resulting energy and anti-gravitational effects would further accelerate the expansion of the universe to create the large-scale smoothing and uniformity that we observe today.



The Big Bang is what we call a singularity event, i.e. a point where the changes in conditions are so infinite and irreversible that all rules of "reality" are changed. In less than a second, the universe had exploded to millions and millions of its original radius. The Big Bang can be considered the beginning of time, since the laws of physics (and even the concept of time itself) only started with it. All prior events fall in the "elsewhere" of our light cone and are both unobservable and inconsequential.



Will our universe expand indefinitely? If space-time began with the Big Bang singularity, it could also end with a singularity, be it (i) a big crunch singularity (where the whole universe re-collapsed), or (ii) a singularity inside a black hole (specifically for the object that fell in). **Current evidence suggests that our universe is expanding 5-10% every 1000mil years, and is likely to expand forever.** Even if it were to re-collapse, it won't happen for at least 10,000 mil years.



Black Holes



How are black holes formed? Stars are created by clusters of free-floating hydrogen atoms brought together by gravity. The atoms interact to produce nuclear reactions, leading to the formation of other elements (mainly helium).

KEY QUOTES

"Time had a beginning at the big bang."

- Stars have a huge gravitational pull. They remain stable when the outward pressure from the core nuclear reactions is balanced with the inward gravitational pressure.
- A star will eventually exhaust its supply of fuel. As it starts to cool and contract, its fate depends on the Chandrasechar Limit:
 - (i) If a star has $<1.5x$ the mass of our sun (i.e. it's *below* the limit), it'll shrink to a fraction of its original mass and become what's known as a white dwarf or neutron star.
 - (ii) If a star is *above* the limit, it'll become too dense to withstand its own gravitational force, thus collapsing to become a black hole.

 Black holes have such infinite gravity that even light can't escape. So, objects can enter but nothing can exit. Inside a black hole, normal scientific laws break down just like the conditions prior to the Big Bang. "Time" and "space" have no meaning.

- It was originally believed that black holes are invisible and can only be detected via their gravitational forces. Later discoveries show that they actually glow like hot bodies due to particles (x-rays and gamma rays) escaping from the area around them.
- Earlier, we explained the existence of particles (with a positive charge) and antiparticles (with a negative charge) which would normally annihilate each other. These also exist in the spaces around black holes. However, the gravity near a black hole is so immense that a negative-energy particle can fall in and get transformed into a positive-energy particle. This allows the partner particle to "survive" and escape the area. Black hole regions emit so much energy that 1 black hole alone could meet our world's electricity needs—if we could find a way to harness it.

 While there's no conclusive evidence of black holes yet, scientists have (i) identified possible black holes in the observable universe and (ii) proven the existence of neutron stars (which have similarly high density). In the meantime,

KEY QUOTES

"Black holes are one of only a fairly small number of cases in the history of science in which a theory was developed in great detail as a mathematical model before there was any evidence from observations that it was correct."

black holes' proven radiation emission have major implications for science and metaphysics. Any object that falls into the black hole is emitted as radiation, which means that matter is never truly destroyed but recycled as energy. A human who fell into a black hole may die, but his mass or energy will still continue to exist in some other form.



Why Backward Time Travel is Unlikely



We've established that time is relative. The next question is whether time moves in a certain direction and if it can travel backward. In imaginary time, you can move forward or backward without any difference. But in "real" time, the direction of time matters.



An arrow of time is what differentiates the past from the future to give direction to time. Hawking believes that backward time travel is unlikely due to 3 arrows of time.

- The Thermodynamic Arrow: According to the second law of thermodynamics, entropy or disorder in a closed system can only increase or stay constant—it can't decrease. Moving time backward means that a broken cup can be made whole again, which violates the law.
- The Psychological Arrow: Psychologically, we can only remember events in the past, not the future. We experience time from a known past to an unknown future with increasing entropy.
- The Cosmological Arrow: Entropy increases as the universe expands (not as it contracts), again implying that time must move forward, not backward.



There are also inherent paradoxes to backward time travel.

- To travel back in time, you must be able to travel faster than the speed of light. Imagine if you were traveling from Point A to Point B. If you moved so fast that you arrive at Point B before the light waves arrive, you can observe your past self leaving Point A. Yet, we've established that it isn't possible to travel faster than the speed of light.

KEY QUOTES

"The laws of science do not distinguish between the forward and backward directions of time. However, there are at least three arrows of time that do distinguish the past from the future."

"The progress of the human race in understanding the universe has established a small corner of order in an increasingly disordered universe."

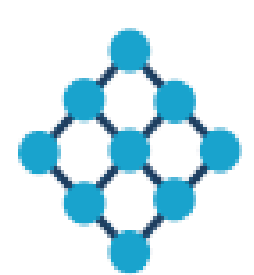
- Another paradox is the “grandfather paradox”. If you could go back in time to kill your grandfather, you won’t exist in the present, which means you couldn’t have gone back to kill your grandfather, and so on. This can only be reconciled if backward time travel doesn’t allow you to alter past events, i.e. you enter an alternate history that doesn’t affect the current space-time.



For now, wormholes offer the best possibility for time travel. The no-boundary proposal says that at the start of space-time, the universe had a warped curvature that allowed time to travel in any direction. This no longer seems possible after the universe expanded and flattened, but perhaps such curvatures still exist in certain areas. If so, quantum physics says that it’s possible to bend space-time to let a wormhole cut through 2 regions of space-time.



The Quest for a Unified Theory



We’ve learned a lot about our universe within a relatively short period of human history. However, we still have some way to go before we have a unified theory that can explain the entire universe. The laws of general relativity don’t work at sub-atomic levels and for singularity events. Quantum mechanics, on the other hand, work for all scenarios but we don’t have a consistent theory that incorporates it with gravity. So, a unified theory must somehow combine the laws of general relativity and quantum mechanics.



So far, the string theory comes closest to bridging this gap.

- The theory basically replaces particles with strings—instead of a particle in a point in space at a point in time, we have strings in a line in space at each point in time. So, every point corresponds to a time and to a position on the string. Waves move up and down the string (similar to how particles collide with one another), to cause strings to combine or split apart.
- String theories seem to describe and predict our universe better than other theories. However, they need up to 26 dimensions of space-time to work. This contradicts

KEY QUOTES

“A complete, consistent, unified theory is only the first step: our goal is a complete understanding of the events around us, and of our own existence.”

our longstanding notion of a four-dimensional space-time. It's possible that we don't see the extra dimensions because they're too tiny to be observable on a large scale, or because we only exist in the regions of the universe with four dimensions.

 As explained earlier, we can never prove a theory 100%. So, we might never know if we've discovered the unified theory. We can only be confident that our theory consistently make predictions that're in line with our observations. Just like how the general relativity and quantum theories are now widely accessible to the masses, **the discovery of a unified theory will allow mankind to completely grasp the laws that govern our universe. Only then will our quest for understanding finally be fulfilled.**

KEY QUOTES

"If we do discover a complete theory...Then we shall all, philosophers, scientists, and just ordinary people, be able to take part in the discussion of the question of why it is that we and the universe exist."